Practical no. 1(a)

Aim: Write the program for the following:
a. Create an MVC application to perform basic Arithmetic Operations.

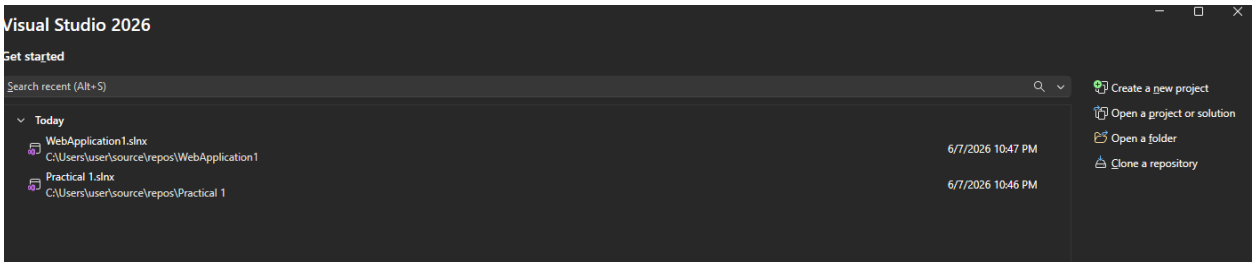
Software Required:

Visual Studio 2026

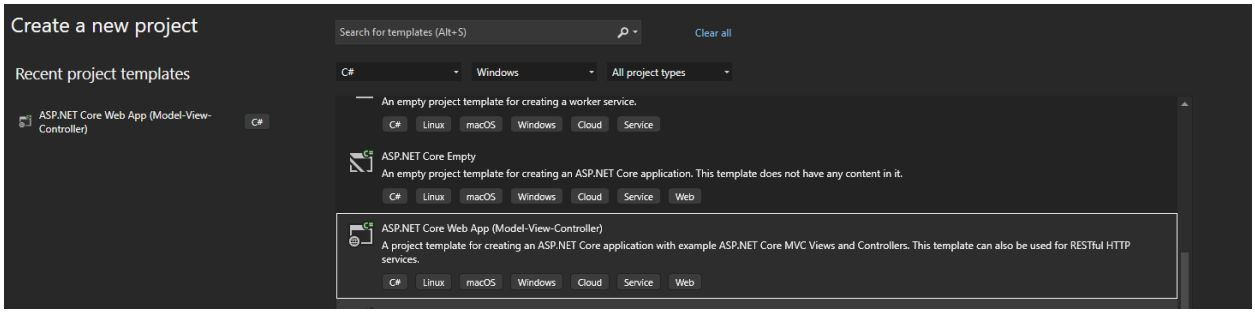
Procedure:

Step 1: Create New MVC Project

1. Open Visual Studio 2022.
2. Click Create a New Project.



3. Select ASP.NET Core Web App (Model-View-Controller).



4. Click Next.
5. Enter Project Name: ArithmeticMVC
6. Click Next.
7. Select .NET 8.0.
8. Click Create.

Step 2: Create Model

1. Click View.

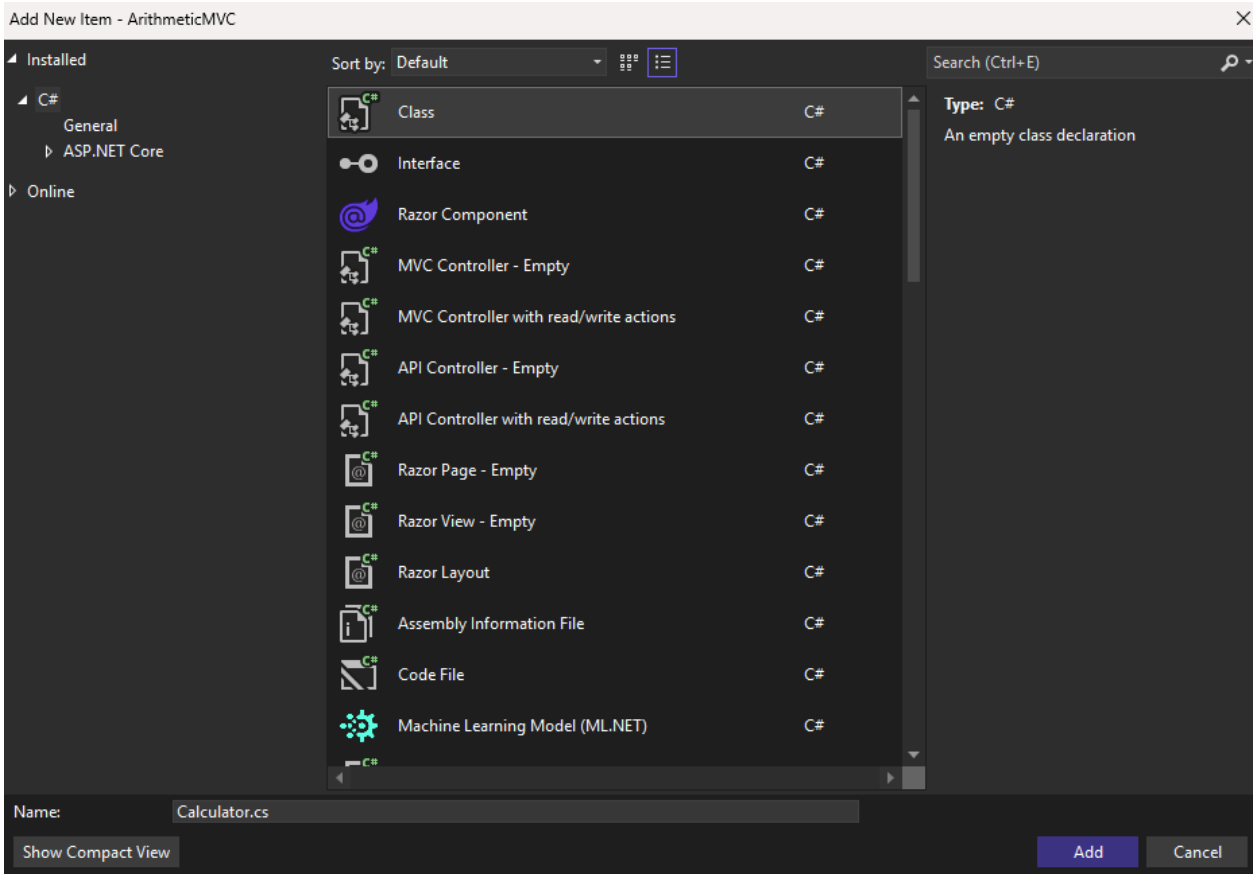
2. Click Solution Explorer..

Create a class named Calculator inside the Models folder.

1. Right Click Models.
2. Click Add.
3. Click New item.
4. Select Class.
5. Give Name Calculator.cs.
6. Click Add.

Calculator.cs

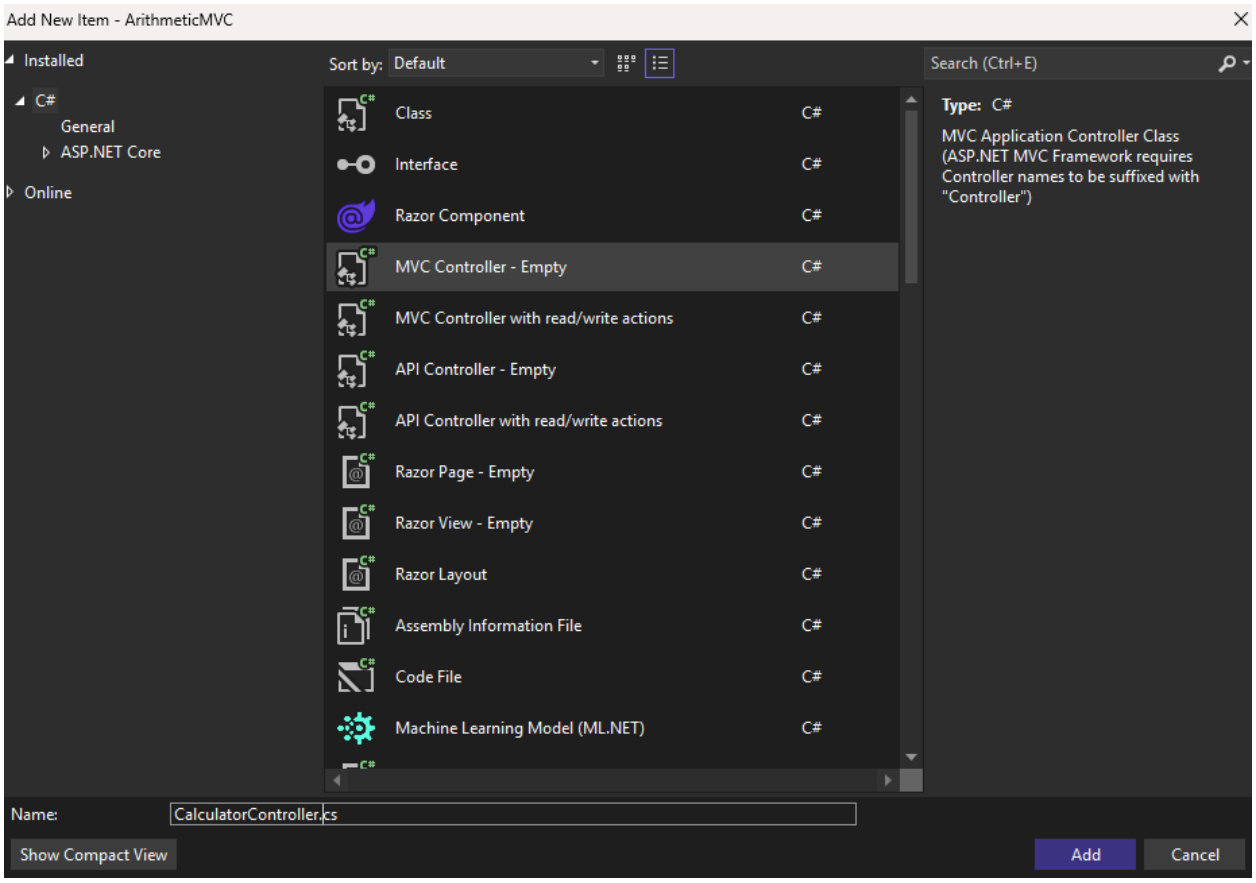
```
namespace ArithmeticMVC.Models
{
   public class Calculator
   {
      public double Number1 { get; set; }
      public double Number2 { get; set; }
      public string Operation { get; set; }
      public double Result { get; set; }
   }
}
```



Step 3: Create Controller

Create a CalculatorController inside the Controllers folder.

1. Right Click Controllers.
2. Click Add.
3. Click New item.
4. Select MVC Controller - Empty.
5. Give Name CalculatorController.
6. Click Add.



CalculatorController.cs

```
using Microsoft.AspNetCore.Mvc;
using ArithmeticMVC.Models;

namespace ArithmeticMVC.Controllers
{
   public class CalculatorController : Controller
   {
      public IActionResult Index()
      {
         return View();
      }

      [HttpPost]
      public IActionResult Index(Calculator calc)
      {
         switch (calc.Operation)
         {
            case "+":
```

```
                calc.Result = calc.Number1 + calc.Number2;
                break;

            case "-":
                calc.Result = calc.Number1 - calc.Number2;
                break;

            case "*":
                calc.Result = calc.Number1 * calc.Number2;
                break;

            case "/":
                if (calc.Number2 != 0)
                    calc.Result = calc.Number1 / calc.Number2;
                else
                    ViewBag.Message = "Division by zero is not allowed.";
                break;
        }

        return View(calc);
    }
  }
}
```

Step 4: Create View

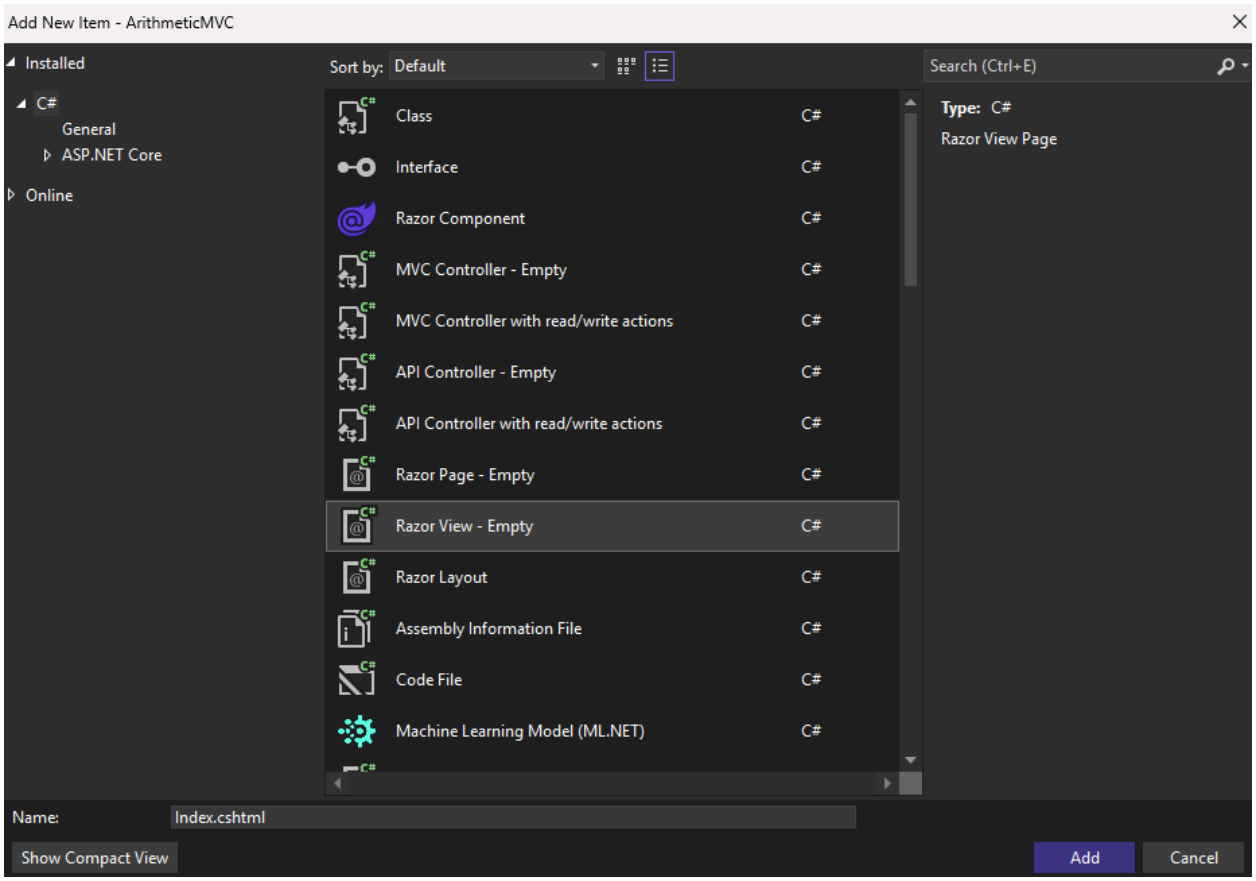
Create the following folder structure:

```
Views
 └── Calculator
     └── Index.cshtml
```

Steps to Create View

1. Right-click Views folder.
2. Select Add → New Folder.
3. Name the folder Calculator.
4. Right-click Calculator folder.
5. Select Add → New Item.
6. Select Razor View.
7. Name it Index.cshtml.

8. Click Add.



Step 5: Add View Code

Index.cshtml

```
@model ArithmeticMVC.Models.Calculator

@{
    ViewData["Title"] = "Calculator";
}

<h2>Arithmetic Calculator</h2>

<form method="post">

    <div>
        <label>First Number:</label>
        <input asp-for="Number1" />
    </div>
```

```
    <br />

    <div>
        <label>Second Number:</label>
        <input asp-for="Number2" />
    </div>

    <br />

    <div>
        <label>Operation:</label>

        <select asp-for="Operation">
            <option value="+">Addition</option>
            <option value="-">Subtraction</option>
            <option value="*">Multiplication</option>
            <option value="/">Division</option>
        </select>
    </div>

    <br />

    <button type="submit">Calculate</button>

</form>

@if (Model != null)
{
    <h3>Result = @Model.Result</h3>
}

@if (ViewBag.Message != null)
{
    <h3 style="color:red">@ViewBag.Message</h3>
}
```

Step 6: Open Program.cs file

Solution Explorer → Program.cs → Change MapControllerRoute.

Program.cs

```
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.
builder.Services.AddControllersWithViews();

var app = builder.Build();

// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    // The default HSTS value is 30 days. You may want to change this for production scenarios,
see https://aka.ms/aspnetcore-hsts.
    app.UseHsts();
}

app.UseHttpsRedirection();
app.UseRouting();

app.UseAuthorization();

app.MapStaticAssets();

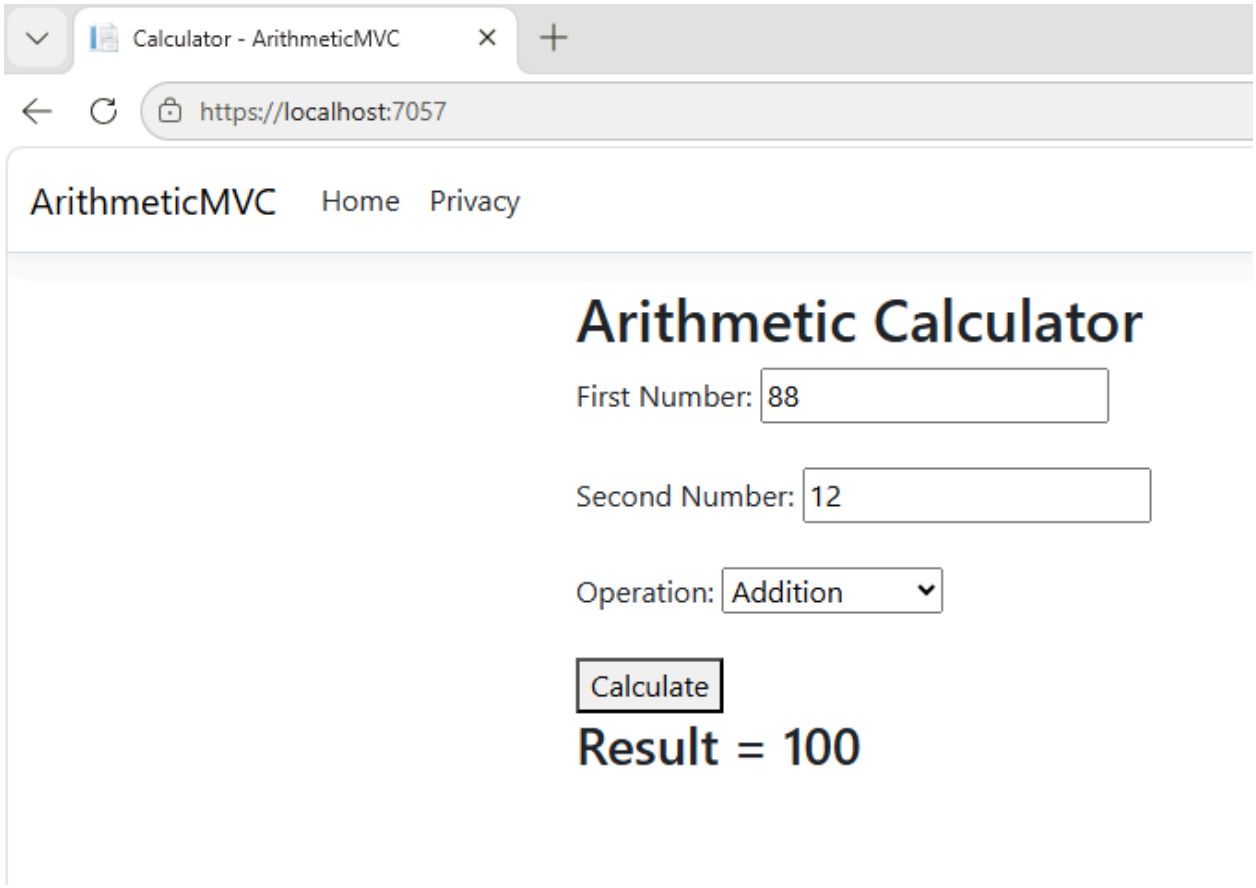
app.MapControllerRoute(
    name: "default",
    pattern: "{controller=calculator}/{action=Index}/{id?}")
    .WithStaticAssets();


app.Run();
```

Step 7: Run Application

1. Click Build → Build Solution.
2. Press F5.
3. The browser opens automatically.
4. Enter two numbers.
5. Select operation.
6. Click Calculate.

Output:


Calculator - ArithmeticMVC
https://localhost:7057
ArithmeticMVC
Home
Privacy
Arithmetic Calculator
First Number: 88
Second Number: 12
Operation: Addition
Calculate
Result = 100